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RESEARCH ARTICLE



The cost of achieving South Africa's 'fair share' of global climate change mitigation

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ABSTRACT

Global climate change mitigation action is hampered by systematic under-assessment of national 'fair shares', largely on the basis of perceived national interests. This paper aims to inform discussions centred on South Africa's nationally determined contribution (NDC) by estimating (1) emissions reduction pathways for the country using the Climate Equity Reference Calculator (CERC) assuming a maximum 2°C aggregate warming target and (2) the likely economy-wide net mitigation costs or savings associated with reaching these pathways if known lower-cost mitigation measures, identified through the national mitigation potential analysis, are prioritised. The cumulative net savings associated with achieving the CERC 'fair share' emissions pathway, assuming the moderate use of low carbon power generation measures, would reach \$5.3 billion by 2030. Net savings could be substantially greater reaching \$46.8 billion by 2030 assuming power generation focuses on moving towards full decarbonisation. An unconditional commitment to the mitigation action implied by the 'fair share' emissions pathway therefore seems reasonable and prudent purely from the point of view of net country-wide savings. Only if power generation moves towards full decarbonisation would there be a reasonable chance of achieving the more ambitious CERC domestic emissions pathway. However, the significant additional cost associated with achieving the domestic emissions pathway should be conditional on international assistance.

Key policy insights

- South Africa can only achieve its 'fair share' of the global mitigation effort if greater use is made of renewable energy options, and can realise significant net savings if it does so.
- Further emissions reductions would incur costs and require significant upscaling of the share of renewable energy and full implementation of all non-power generation mitigation measures available.
- Committing to this further mitigation action contingent on international finance would both strengthen the nation's position in climate negotiations and support the provision of finance for those vulnerable developing nations that bear little or no responsibility for climate change.

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Introduction

Achieving global greenhouse gas emissions reduction goals is contingent on the implementation of mitigation by all countries, but the assignment of 'fair shares' of action is challenging and contentious. The Civil Society Equity review represents a consensus position from a significant number of non-governmental organisations on the assessment of 'fair shares' in the hope of pressurising nations to undertake action (CSO Review, 2015),

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and this paper follows the same methodology. Since most countries in the developed world have a large historical responsibility and high capacity levels, part of their 'fair share' of action must consist of supporting low carbon development in developing countries that have to reduce emissions beyond their own 'fair share' (CSO Review, 2015; den Elzen & Höhne, 2010; Höhne, den Elzen, & Escalante, 2013; Holz, Kartha, & Athanasiou, 2017; Winkler, Letete, & Marquard, 2013). This is recognised in climate finance commitments under the United Nations Framework Convention on Climate Change (UNFCCC). An effort-sharing approach to estimating 'fair shares' divides the necessary global mitigation effort according to each individual nation's needs, capability and responsibility¹. The magnitude of necessary mitigation and its associated costs should thus be an important informant of decision-making, and many countries, including South Africa, have taken steps towards enhancing their understanding in this regard. This includes the drafting in 2014/2015 of Intended Nationally Determined Contributions (INDCs) to the global mitigation effort as articulated in the 2015 Paris Agreement, which, subsequent to ratification of the Agreement, have become NDCs. These voluntary country commitments reflect a country's view on what their mitigation and adaptation contributions should be, based on national circumstances and development objectives. In 2018, Parties to the UNFCCC will engage in a Facilitative Dialogue – now known as the Talanoa Dialogue – at the 24th Conference of the Parties (COP 24) focused on taking stock of current mitigation opportunities and enhancing ambition in the NDCs. This article aims to inform discussions in this regard by estimating emissions reduction targets for South Africa, along with the likely net mitigation costs associated with reaching them. The basic point of departure for these estimations is a global target limiting climate change to 2°C, and the requirement to achieve a national 'fair share' of emissions effort. This differs from the 'well below' 2°C target of the Paris Agreement, but it does represent a level of ambition that improves on current NDC commitments, and has long been accepted as reasonable within the climate negotiations space. Moreover, it aligns with many prior fair share assessments (Winkler et al., 2013) and the South African commitment within the NDC.

It is worth noting that for the first batch of NDCs, there was limited agreement between countries on what to include and how to present key data (Overseas Development Institute [ODI], 2015) – a relatively unsurprising outcome given they were a new concept, their voluntary nature and the likelihood that information would be provided sparingly as part of strategic behaviour ahead of climate change negotiations. In particular, very few NDCs included information on the costs of mitigation. The South African NDC was one of relatively few that included any data on mitigation costs and only provided some fairly dated and highly preliminary high-level data on gross mitigation costs (i.e. excluding potential cost savings) based on the 2007 Long-Term Mitigation Scenarios (LTMS). The UNFCCC provides broad guidance on what INDCs should contain, but, detailed methodological guidance has not been negotiated by the Parties. The UNFCCC secretariat website provides links to guidance which countries may wish to consult including the Climate and Development Knowledge Network (CDKN)'s Guide to INDCs (CDKN, 2015), the International Partnership on Mitigation and MRV (IPMM) INDC guidance document (IPMM, 2014), the World Resources Institute (WRI)/United Nations Development Programme (UNDP) INDC Guide (WRI, 2015a) and the South Centre Policy Brief on INDCs (South Centre, 2015). These guidance documents were reviewed for this paper in terms of their specific consideration of how best to treat mitigation costs in NDCs. It was found that the IPMM document provides the most detailed guidance in this regard. It recommends that such costs should be compared to a business-as-usual (BAU) scenario. In addition, the availability of lower-cost measures and those that result in cost savings (when lower power or fuel costs are taken into account, for example) should be considered when evaluating whether contributions are ambitious (IPMM, 2014). This conceptual approach is followed in this article to estimate the costs associated with South Africa's emissions pathways and the evaluation of these in the context of contributions to global mitigation efforts.

In the remainder of the article, context is provided regarding South Africa's contribution to historical emissions along with its official emissions targets. The Climate Equity Reference Calculator (CERC) is then used to estimate future emissions pathways to 2030 for South Africa associated with varying levels of ambition and commitment from the international community. This is followed by estimates of the country-wide net costs or savings associated with achieving an indicative 'fair share' of global emissions reductions. These estimates are then discussed in terms of their implications for the country's unconditional and conditional commitments.

Emissions pathways for South Africa

South Africa contributes just over 1% of global emissions and is ranked 14th out of 185 countries in terms of its historical contribution to cumulative CO₂ emissions between 1850 and 2013 (Janssens-Maenhout et al., 2017; WRI, 2015b). The latest draft GHG inventory (Department of Environmental Affairs [DEA], 2016) reports total emissions (excluding land use) in 2012 of 544.6 MtCO₂e – an increase of 20.9% from emissions in 2000 and 57% from 1990 levels. Other estimates of national emissions range from 451 MtCO₂e to 493 MtCO₂e in the same year (Janssens-Maenhout et al., 2017; WRI, 2015b). Compared to other middle-income countries, South Africa's per capita emissions are relatively high at 8.09 tCO₂ per capita (40th out of 181 countries) (WRI, 2015b). This can be attributed to South Africa's continued reliance on coal as an energy source in the energy sector as well as in industry (International Energy Agency [IEA], 2017).

South Africa's emissions trajectory and targets

South Africa has committed to contributing its 'fair share' to global efforts to reduce greenhouse gas emissions (DEA, 2010). In late 2009, the President announced the intention to implement mitigation actions that would result in a 34% and 42% deviation from a BAU emissions growth trajectory by 2020 and 2025, respectively, plateau for a decade, and decline thereafter. This pledge did not specify the BAU trajectory and was thus opaque in terms of the actual emissions trajectory proposed. This 'peak-plateau-decline' (PPD) trajectory was formalised in the National Climate Change Response Policy (DEA, 2011a) as the national benchmark trajectory range where the following emission targets were specified:

- Peak: Emissions peak in a range with a lower limit of 398 MtCO₂e and upper limits of 583 and 614 MtCO₂e in 2020 and 2025, respectively.
- Plateau: Emissions plateau for up to 10 years and remain within the 398 and 614 MtCO₂e range.
- Decline: From 2036 or before, emissions decline to a range between 212 MtCO₂e and 428 MtCO₂e in 2050.

South Africa's NDC reflects the national benchmark trajectory range to 2030, aligning with the NDC time horizon. While the NDC target provides more clarity than the 2009 pledge as it is quantified, the target has come under criticism as the wide range does not provide any certainty as to South Africa's intended emissions pathway range (Climate Action Tracker [CAT], 2017; Worldwide Fund for Nature [WWF], 2015), and that it falls short on ambition (Rocha et al., 2015).

The NDC commits to these targets conditional upon 'the effective implementation by developed country Parties of their commitments under the Convention relating to financial resources, development and transfer of technology, and capacity building' without specifying what it considers would represent such effective implementation (DEA, 2015, p. 7). It also considers 'the PPD range to be an equitable contribution to the global mitigation effort, given South Africa's current and historical emissions and its national circumstances (especially its development challenges)' whilst acknowledging that most independent assessments of a 'fair share' would require considerably more ambition (DEA, 2015, p. 10).

A differentiation, therefore, between a reasonable 'fair share' for the country and a specific target that aligns with limiting climate change to no more than 2°C is useful for analysing the relative costing of required national action, and quantification of the necessary international support for further action. This would provide a better delineation for conditional commitments, with an unconditional component corresponding to South Africa's 'fair share', and proposed additional mitigation action conditional on the provision of climate finance.

Climate equity and determining 'fair shares' of global mitigation efforts

The determination or allocations of such 'fair shares' is a famously contested space in climate change policy negotiations. Whilst the UNFCCC includes the term 'common but differentiated responsibilities and respective capabilities' (UNFCCC, 1992), the actual allocation of responsibilities and differentiated goals has historically proved a sticking point in the UNFCCC process, and is one of the main reasons for the current bottom-up

NDC approach articulated in the Paris Agreement. Nevertheless, to align with realistic carbon budgets that correspond to global climate mitigation targets, a top-down assessment of allocations is required. Moreover, since carbon budgets are essentially zero sum, the allocation and achievement of 'fair shares' necessarily imply political and financial exchanges between countries (Winkler et al., 2017; Winkler, Vorster, & Marquard, 2009).

A number of approaches and tools are available for assessing 'fair shares' (Baer, Fieldman, Athanasiou, & Kartha, 2008; CAT, 2017; du Pont, Jeffery, Gütschow, Christoff, & Meinshausen, 2016; Pan, den Elzen, Höhne, Teng, & Wang, 2017), and it is notable that quantified approaches have better traction in the literature. The Climate Equity Reference Project (CERP) (Holz et al., 2017) undertakes such a 'fair share' allocation through the assessment of a level of 'development need' together with indicators of 'responsibility' and 'capability', which consider factors such as historical emissions, GDP per capita and income distribution.

Justifying a 'fair share' requires detailed clarification of the input assumptions, since there is no division of mitigation action that can align with all parties' positions. It is important therefore to note that the use of a common tool simplifies this process, opening all assumptions to interrogation. By making use of such a tool a country can therefore meet the Lima Call for Climate Action issued at COP 20 in 2014 to explain 'how the Party considers that its intended nationally determined contribution is fair and ambitious' (UNFCCC decision 1/CP.20 para 14). This justification has typically been poor in current NDCs (Winkler et al., 2017), and making use of such a tool could improve the rigour and interrogability of such justifications.

The CERP² has developed such a systematic tool that enabled civil society to undertake a rapid analysis of the adequacy of early INDC submissions in the lead up to the Paris Agreement (CSO Review, 2015). The CERC is a tool that allows users to transparently explore 'fair share' calculations by specifying a selection of key parameters. The outcomes of the Calculator are country or region reports, which detail the baseline emissions, the calculated 'fair share' of the global mitigation requirement as well as an indicative domestic emissions pathway. The domestic emissions pathway is seen to decline (relative to the national baseline) at the same rate that global emissions would need to decline below the global baseline to ensure limiting of global climate change to the target temperature.

The CERC notes the following caveat in this regard:

In the real world, a national domestic emissions trajectory will depend on the cost of domestic mitigation relative to the cost of mitigation in other countries, and on its chosen participation in international mechanisms for providing or receiving financial and technological support for mitigation. (Kemp-Benedict, Holz, Baer, Athanasiou, & Kartha, 2017)

This means that the domestic emissions trajectory is indicative rather than prescriptive. However, for the purposes of this study, the domestic emissions trajectory was used as a target in estimating the further mitigation potential, and to evaluate the extent to which international climate finance is required to provide further mitigation in the South African context.

The CERC was used in this paper to estimate 'fair share' emissions and domestic emissions pathways for South Africa to 2030.³ This is not intended to be a prescriptive scenario, but rather an indicative one, to allow for exploration of the implications of a 'fair share' commitment. This paper takes a 'middle of the road scenario', corresponding to the '1950/medium-progressivity' equity benchmark described in the Holz et al. (2017) assessment of global 'fair shares'. This scenario is specified by an equal balancing of capability and responsibility (derived from cumulative emissions since 1950), a 'medium progressivity' benchmark (which excludes a 'development threshold' of \$7500 below which an individual's income is considered exempt), and ensuring a 66% chance of limiting global warming to no more than 2°C⁴ over the long term. The scenario mostly aligns with the middle of the Civil Society Review (2015)'s benchmarks and was selected to represent a compromise position between common contending positions.⁵ The one point of difference is in the level of global ambition selected, with this work using 2°C and the Civil Society Review preferring 1.5°C, with the view that a greater than or equal to 50% chance of being below 1.5°C is closer to a '2°C pathway' according to IPCC definitions (CSO Review, 2015).

Figure 1 shows the key results of applying the calculator to a global target that limits temperature increases to 2°C. South Africa's CERC emissions baseline⁶ without additional measures is projected to reach 758 MtCO₂e per year by 2030. Although this differs from national baseline estimates specified in various government

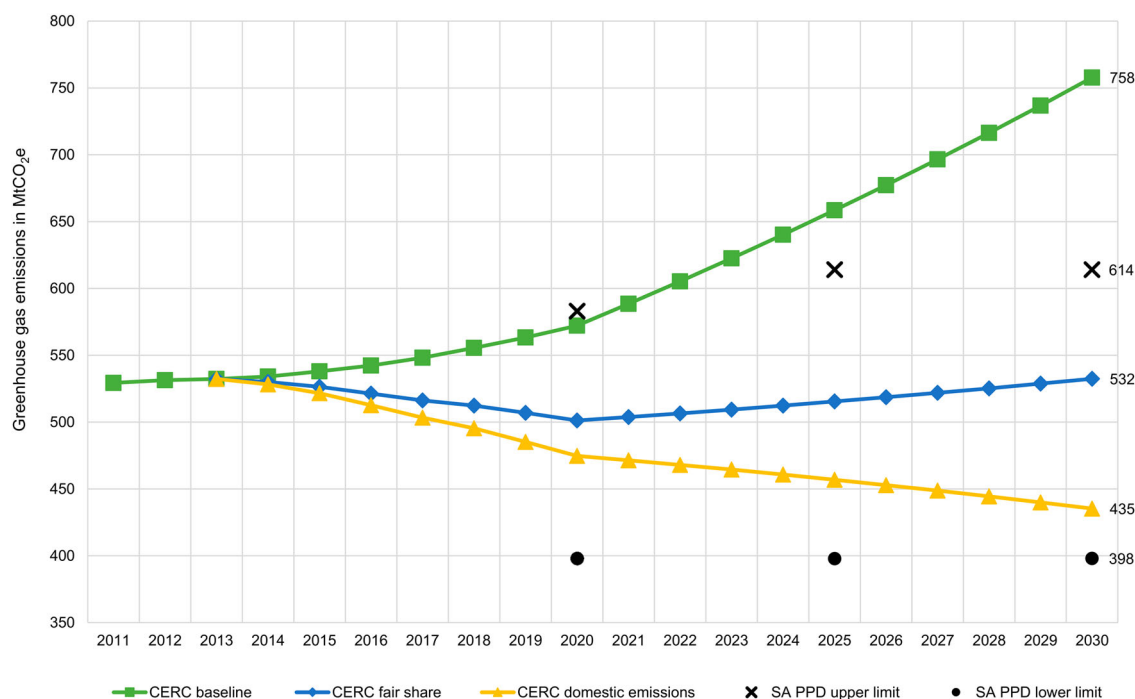


Figure 1. Climate equity reference calculator (CERC) emissions pathways for South Africa with global target to limit temperature increase to 2°C.

documents (DEA, 2014; Winkler, 2007) and other studies (Altieri et al., 2016), it does fall within the baseline that was used to calculate the benchmark emission trajectory range. The benchmark trajectory range was calculated using an upper and lower baseline or ‘business-as-usual’ limit, which ranged from 755 MtCO₂e to 1255 MtCO₂e per year by 2030 (DEA, 2011b). Thus, the CERC baseline corresponds to the baseline used to calculate the lower ‘peak-plateau-decline’ (PPD) trajectory and as such was utilised instead of attempting to adjust baselines. Furthermore, by making use of consistent data sets for all countries, the CERC provides a more consistent means of comparing ‘fair shares’. The reductions required to achieve the CERC ‘fair share’ pathway, relative to the CERC baseline, would have to gradually build to 71 MtCO₂e in 2020, 143 MtCO₂e in 2025 and reach 226 MtCO₂e in 2030 (i.e. 758 MtCO₂e–532 MtCO₂e). The additional reductions required to reach the CERC domestic emissions pathway, relative to the ‘fair share’ emissions pathway, would build to 97 MtCO₂e by 2030 (i.e. 532 MtCO₂e–435 MtCO₂e). For comparative purposes, Figure 1 also shows the upper and lower limit of the South African PPD trajectory.

Note that South Africa’s ‘fair share’ pathway reflects the level of mitigation required that takes into account both the nation’s historical responsibility for greenhouse gas emissions and its current ability to mitigate. Interpreting this in light of national commitments, the ‘fair share’ pathway is most consistent with South Africa’s ‘unconditional commitment’. The balance of additional mitigation effort that would be required to achieve the domestic emissions pathway is then most consistent with the country’s ‘conditional commitment’ – additional action required to achieve the global target that South Africa might undertake, providing that international support is provided.

Methodology and data sources for estimating mitigation costs or savings

The analysis of the mitigation costs or savings associated with achieving the CERC fair share emissions pathway relied on existing data generated by the South Africa Department of Environmental Affairs (DEA) and the Council for Scientific and Industrial Research (CSIR), the premier parastatal research and development institute in the

country. Significant DEA initiatives focused on mitigation costs include the LTMS finalised in 2007, which provided data on mitigation potential, costs and benefits. The LTMS was a key informant of the National Climate Change Response Policy completed in 2011. The updated Mitigation Potential Analysis (MPA) was then completed in 2014 and included relatively detailed cost estimates for 175 mitigation measures (see DEA, 2014). These MPA cost estimates were the primary source of data for this assessment of the mitigation costs associated with emissions reductions. In addition, it was possible to update the costs for mitigation associated with power generation contained in the MPA (which drew on dated Integrated Resource Plan (IRP) 2010 data) so that they reflected more recent Draft IRP 2016 data along with that contained in a review of the Draft IRP 2016 conducted by the CSIR (see CSIR, 2017).

The approach used for estimating the net annual country-wide cost (or savings) of achieving South Africa's 'fair share' of mitigation involved the following key steps:

- (1) Estimate the costs or savings of all power generation mitigation measures based on CSIR power generation scenarios relative to the Draft IRP 2016 Base Case. The inclusion of all power generation measures was considered the most realistic approach as South Africa is set to invest heavily in the energy sector as a prerequisite of development, as well as to remedy the current ageing generation infrastructure. Considering that, in 2012, electricity generation was responsible for over 45% of total national emissions (DEA, 2016), it is clear that the principal mitigation gains are to be won in this sector. Moreover, since the source data (CSIR, 2017) analyses full system development, it was not possible to isolate individual power generation measures.
- (2) For all non-power generation mitigation measures contained in the MPA, estimate their net costs (i.e. costs minus potential savings when lower power or fuel costs are taken into account) per volume of emissions abated between 2017 and 2030 based on MPA data appropriately inflated.
- (3) Sort or rank all non-power generation mitigation measures from lowest net cost (or highest saving) to highest net cost as reflected by their net costs per volume of emissions abated between 2017 and 2030.⁷
- (4) Starting with the lowest net cost (or highest saving) measure, identify all the non-power generation measures whose emission reductions, when added to those associated with power generation measures, would sum to the annual emissions reduction required to achieve the CERC 'fair share' pathway.
- (5) Sum the net annual costs (or savings) of all the power generation and identified non-power generation mitigation measures to establish the annual net costs (or savings) of all mitigation measures required to achieve the CERC 'fair share' pathway. Use these annual estimates to generate cumulative estimates for the period 2017–2030.

The cost implications of relaxing the assumption of 100% implementation or uptake of non-power generation measures were also explored, given the greater uncertainty associated with these measures. All estimates were initially estimated in 2017 South African Rand and converted to United States Dollars at prevailing exchange rates.⁸ The following sections elaborate on the data sources and assumptions used for estimating the costs or savings associated with mitigation measures making the distinction between those in the power generation sector and non-power generation measures.

The costs or savings of power generation mitigation measures

For the overall energy sector, the MPA notes that power generation or electricity supply accounted for 65% of all energy-related emissions in 2009 (DEA, 2014). MPA scenarios with respect to power generation sources and costs were based on planned capacity additions to meet demand and rely on data from the IRP for Electricity 2010–2030 (Department of Energy [DoE], 2011). In keeping with the principles applied in the MPA, the more recent Draft IRP 2016 Base Case Scenario was assumed to align with the With Existing Measures (WEM) scenario for this assessment.⁹ This scenario would entail 68% coal, 22% renewable and 10% other power generation sources (i.e. nuclear, gas and biomass) by 2030.¹⁰ The following With Additional Measures (WAM) scenarios were then assessed relative to the Draft IRP 2016 Base Case (CSIR, 2017):

- (1) The CSIR Least Cost Scenario, which would entail a moderate level of decarbonisation by increasing the share of renewable power generation sources to 34% whilst that of coal decreases to 54% by 2030. It was also found to be the lowest cost scenario tested by the CSIR (hence its name).
- (2) The CSIR Decarbonised Scenario, which would entail a high level of decarbonisation by increasing the share of renewable power generation to 53% whilst that of coal decreases to 36% by 2030.

Figure 2 draws on annual CSIR data to show the differences in greenhouse gas emission per power generation scenario. The CSIR Least Cost Scenario is similar to the IRP 2016 Base Case until 2025, after which it results in gradually lower comparative emissions which build to 47 MtCO₂e less emissions per year by 2030. The CSIR Decarbonised Scenario, which would involve more significant investment in renewable energy sources, results in substantially lower greenhouse gas emissions, reaching 110 MtCO₂e less emissions by 2030 compared to the Draft IRP 2016 Base Case.¹¹

The costs of the CSIR Least Cost Scenario are similar to the Draft IRP 2016 Base Case until 2025 after which the Least Cost Scenario results in gradually lower comparative costs (due to the lower cost of installation of renewable power generation sources when compared to other sources such as nuclear and coal power.) It results in cumulative savings which build to \$3.32 billion by 2030 (equating to 1% of the total power generation costs of the Draft IRP 2016 Base Case over the same period). The Least Cost Scenario thus results in moderately decreased emissions whilst also allowing for some cost savings. The CSIR Decarbonised Scenario would entail greater costs than the Draft IRP 2016 Base Case that would reach a cumulative total of \$0.55 billion by 2030 (equating to 0.16% of the total power generation costs of the Draft IRP 2016 Base Case over the same period). The Decarbonised Scenario thus results in highly significant decreases in emissions at relatively marginal additional costs. The aggressive roll out of renewable power generation options would thus not lead to particularly significant additional costs.

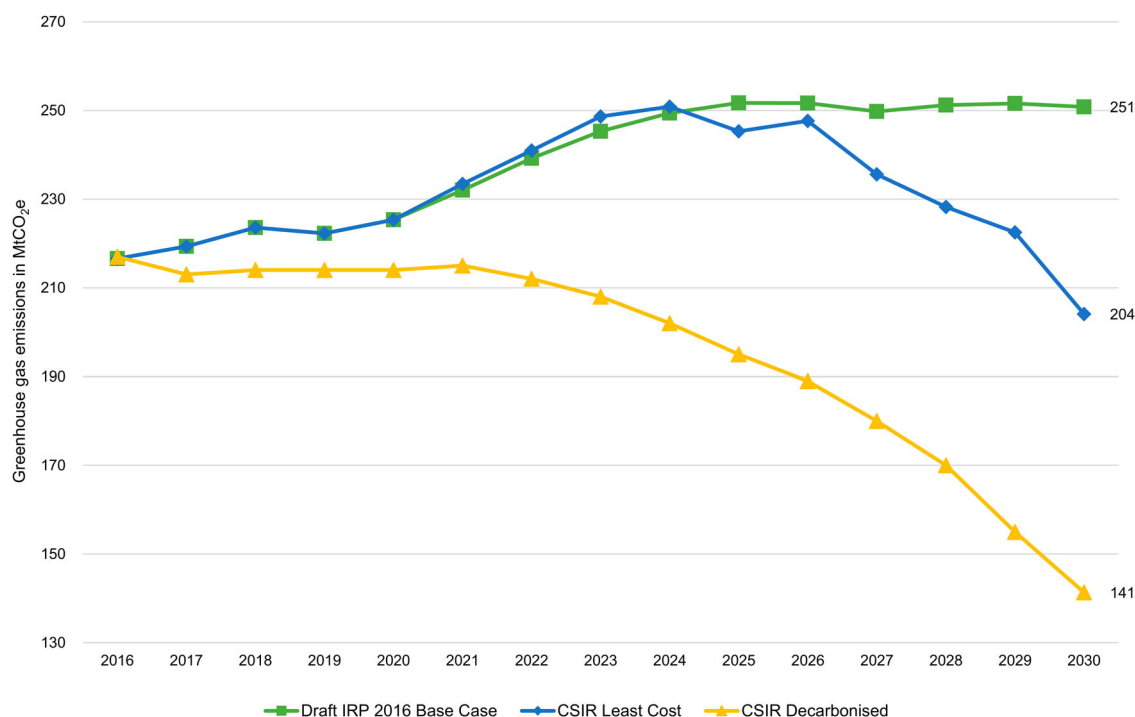


Figure 2. Comparative greenhouse gas emissions per power generation scenario (2016–2030).

The costs or savings for non-power generation mitigation measures

Data on emissions abatement and net costs for non-power generation mitigation measures was sourced exclusively from the MPA, which presented costs in 2010 terms. These were then inflated to arrive at 2017 estimates. Linear projection was used to estimate annual emissions abatement and Marginal Abatement Cost (MAC) amounts not provided in the MPA (for example, data were generally available for 2020 and for 2030 but had to be interpolated for the intervening years). It was assumed that the abatement and net cost estimates in the MPA were largely accurate noting that net costs were negative, therefore implying net savings, for 77 measures out of a total of 165 non-power generation measures. The following caveats with respect to the nature of the MPA and its data also need to be kept in mind:

- The MPA does not represent a comprehensive mapping of all possible mitigation measures for all sectors. Some minor sectors were not included in the analysis, notably downstream manufacturing, and additional measures may exist for sectors that were included in the analysis.
- The MPA data assume that many measures would be implemented to their full technical potential and assumes an uptake trajectory between 2010 and 2050. Both the rate and extent of uptake may not be feasible or fully realised.
- Financing requirements are not considered. For some measures access to finance is less likely to be a constraint. Measures with negative net costs are less likely to struggle in attracting finance though there may be political or social barriers to implementation.

Whilst this assessment focuses exclusively on cost considerations, there are other criteria that could and should inform the availability and choice of measures, particularly in the South African context. These include impacts on jobs, other environmental impacts, relative ease of implementation and regulatory hurdles. The MPA also does not consider the impact of a carbon tax in determining the cost savings associated with mitigation measures. These shortcomings and caveats aside, the MPA still represents the most comprehensive and best available cost analysis of mitigation in the South African context, which is why it was selected to support this analysis.

Results for mitigation costs or savings estimates

For ease of understanding, net annual cost or savings estimates of the mitigation required to achieve the 'fair share' emissions pathway were generated for two overall scenarios named the Low Carbon and Decarbonised scenarios. The former reflects the power generation assumptions associated with the CSIR Least Cost Scenario while the latter reflects those of the CSIR Decarbonised Scenario. Before discussing the results of net cost or savings estimates, it is useful to consider the maximum emissions reduction potential of each scenario. This was done by constructing attainable emissions pathways per scenario when including the power generation measures plus the full implementation of all the non-power generation mitigation measures contained in the MPA. Figure 3 shows the resultant emissions pathways for the scenarios relative to the baseline, 'fair share' and domestic emissions pathways. Both the Low Carbon and the Decarbonised Scenario would allow for the achievement of the CERC 'fair share' emissions pathway. However, only the Decarbonised Scenario stands a reasonable chance of achieving the CERC domestic emissions pathway. The availability of mitigation measures, from among those under consideration here, therefore becomes a constraining factor particularly if the domestic emissions pathway is to be attained. It should be borne in mind that this assumes a 2°C maximum global temperature increase target and greater constraints would apply under a more ambitious 1.5°C maximum temperature increase target.

Figure 4 shows the overall net mitigation costs required to achieve the CERC 'fair share' emissions pathway for each scenario, assuming full implementation or uptake of non-power generation measures from the MPA. The Low Carbon Scenario reduces emissions via power generation measures but to a significantly lesser degree when compared to the Decarbonised Scenario. It therefore requires the inclusion of substantially more non-power generation measures than the Decarbonised Scenario. It would entail overall net negative

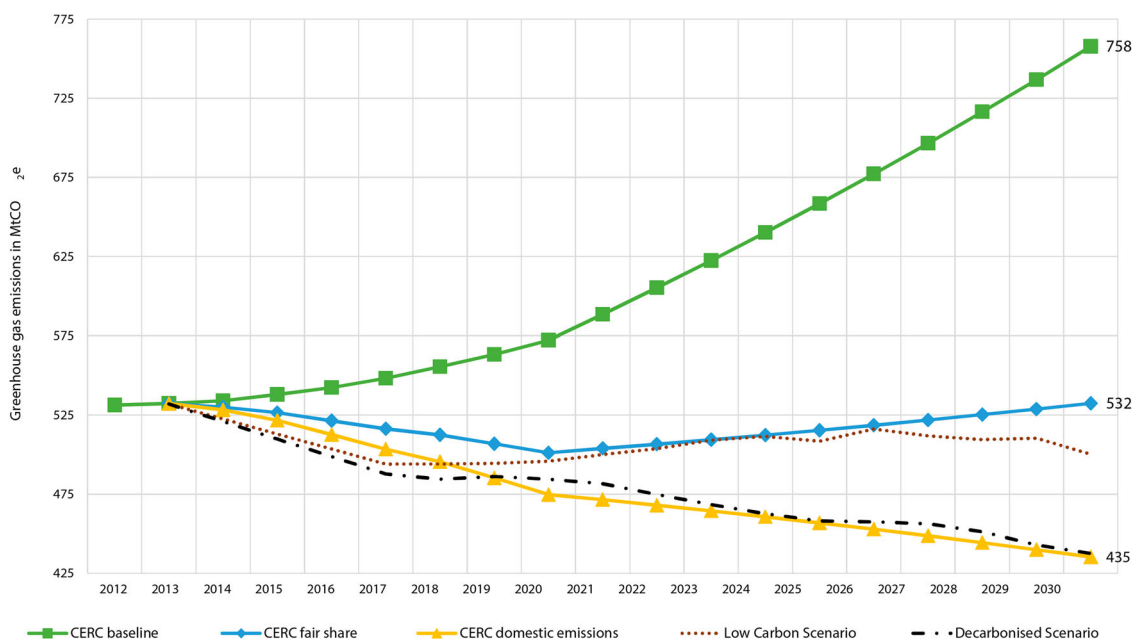


Figure 3. Emissions pathways achievable for the scenarios when including all measures compared to the CERC baseline, ‘fair share’ and domestic emissions pathways.

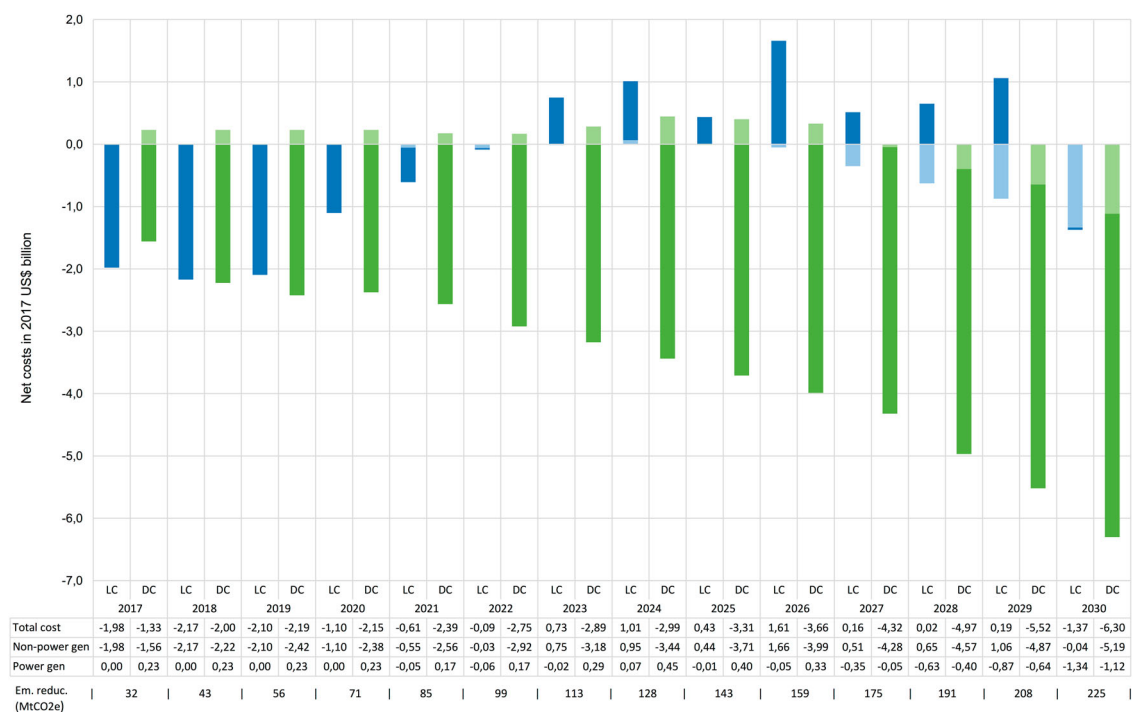


Figure 4. Cost and savings per annum for the achievement of the CERC ‘fair share’ emissions pathway assuming full implementation or uptake of non-power generation measures (LC = low carbon scenario, DC = decarbonised scenario, lighter shaded areas in columns present power generation measures and darker shaded areas are for non-power generation measures, emissions reductions per year specified).

costs (i.e. savings) of \$1.1 billion in 2020, net costs of \$0.43 billion in 2025 and net savings of \$1.37 billion in 2030.¹² Cumulative net savings would reach \$5.3 billion by 2030. The Decarbonised Scenario would result in net savings of \$2.15 billion in 2020, \$3.31 billion in 2025 and \$5.3 billion in 2030. Cumulative net savings would reach \$46.8 billion by 2030 implying that it presents an opportunity for greater savings relative to the Low Carbon Scenario.

The assumption of full or 100% implementation or uptake of MPA non-power generation measures is arguably overly optimistic given likely implementation constraints. This assumption was therefore relaxed to find thresholds below which the emissions reductions required would no longer be possible given the limited number of measures in the MPA (in other words, the point at which there would be insufficient MPA non-power generation measures available to achieve the emission pathways when combined with the power generation measures). For the Low Carbon Scenario, it was found that anything lower than full implementation of non-power generation measures would jeopardise reaching the 'fair share' pathway in more than two years post 2017. Reduced implementation scenarios were therefore not further assessed for the Low Carbon Scenario. For the Decarbonised Scenario, anything lower than 70% implementation of non-power generation measures would jeopardise reaching the 'fair share' pathway. The costs associated with this level of reduced implementation of non-power generation mitigation measures would be higher as it is necessary to include the highest cost measures (i.e. those that appear towards the end of the list of measures ranked from lowest to highest cost). There would be net costs of \$0.19 billion in 2020, net savings of \$0.31 billion in 2025 and net savings of \$2 billion in 2030. Cumulative net savings would reach \$6.1 billion by 2030.

In summary, the assessment of the scenarios shows that South Africa can achieve its 'fair share' of the global mitigation effort with net savings which are maximised when greater use is made of renewable energy options as per the Decarbonised Scenario. An unconditional commitment to the mitigation action implied by the 'fair share' emissions pathway therefore seems reasonable and prudent purely from the point of view of net country-wide savings. It should be borne in mind that short-term costs, much of which would be borne by the state, would still have to be incurred to achieve these longer-term savings. In addition, barriers to implementation are an important consideration though not dealt with in this article. They can vary substantially between sectors that face very different constraints and opportunities. See Spencer et al. (2017) for coal sector transition considerations, for example. The current economic environment, characterised by low growth levels, along with other factors such as the uncertain future of the Renewable Energy Independent Power Producer Procurement Programme (REIPPPP), are also sure to play a role in determining eventual outcomes.

Implications of the results for achieving the CERC domestic emissions pathway

Only the most ambitious mitigation scenario considered, namely the Decarbonised Scenario assuming full implementation of non-power generation measures, stands a reasonable chance of achieving the CERC domestic emissions pathway or coming close to doing so as shown in [Figure 3](#). For this scenario, it can be argued that South Africa could justifiably strive to have the costs of action beyond the 'fair share' covered by international climate finance. Since the least cost mitigation actions will already have been undertaken unilaterally, the cost of additional action is likely to be considerably higher per unit of reduction. Consequently, a prudent negotiating position for South Africa would be to fully quantify the gross costs of action beyond the 'fair share', but commit only to take such action if climate finance is provided.

There are considerable political restraints on the provision of international climate finance (Purdon, 2013), but it is likely that stronger unilateral action and clearer quantification on the part of developing nations can increase moral pressure within the international policy space for increased climate finance provision. The significant net country-wide savings associated with achieving the 'fair share' emissions pathway indicates that some action beyond the fair share could also be achievable at a net saving for the country. Although not the subject of this article, this outcome would benefit from further consideration, along with other pertinent factors, in terms of whether it provides a justification for South Africa to unconditionally commit to going beyond its 'fair share' and to what degree.

Conclusion

This assessment provides an indication of how a developing nation might structure its NDC to clarify and cost both conditional and unconditional action in line with a 'fair share'. Moreover, the use of a tool like the CERC provides enhanced clarity and justification for the 'fair share' assessment within the NDC that can be clearly interrogated by interested parties.

The assessment shows that South Africa can achieve its 'fair share' of the global mitigation effort with significant net savings if greater use is made of less carbon intensive, and often lower cost, renewable energy sources. An unconditional commitment to a 'fair share' of international mitigation action therefore seems justifiable on the basis of this analysis. This will strengthen the nation's moral position when calling for supplementary climate finance for itself and other developing nations from those nations whose 'fair share' exceeds their internal national mitigation potential. Moreover, by improving the South African NDC within the global climate negotiations to reflect this 'fair share' commitment, South Africa can set a critical precedent in the 2018 Talanoa Dialogue to enable the increased global ambition necessary to limit climate change to 2°C.

If the Decarbonised Scenario is selected, it should also be possible to reach the more ambitious CERC domestic emissions pathway with a commitment to the significant upscaling of renewable energy investment. Incurring the costs associated with achieving this pathway should be conditional on international climate finance, since this is the obligation of those nations with a greater historical responsibility.

Interestingly, following the Low Carbon (higher cost) Scenario to achieve a national 'fair share' limits the potential for additional mitigation action because it makes more use of non-power mitigation options. This in turn reduces the potential climate finance that can be spent on mitigation, as well as reducing the likelihood of achieving a global mitigation goal. Purely in terms of mitigation finance and long-term climate goals, then, it is rational for South Africa to pursue the more aggressive Decarbonisation Scenario.

This assessment has focused on emissions pathways and costs to 2030 in alignment with the NDC time horizon. While it is feasible for South Africa to achieve its 'fair share' (and more), the greater mitigation challenge is likely to occur after 2030 when, as per South Africa's commitment, emissions need to start declining in absolute terms. Much more will need to be done for South Africa's rate of emission reduction to increase significantly post 2030. Nevertheless, this is a further motivator to undertake as much cost saving and low-cost mitigation as possible up front, particularly in the power generation sector, which will provide the cheapest mitigation wins assuming coal is not locked in through development in the pre-2030 period.

Notes

1. It is acknowledged that there are other approaches to determining 'fair shares'. However, this is the approach followed in this paper.
2. See: <https://climateequityreference.org>
3. See <https://calculator.climateequityreference.org/>
4. The 66% likelihood is derived from the UNFCCC Fifth Assessment Report and correlates to a total global carbon budget that has a 'better than even' likelihood of limiting climate change to the target temperature.
5. See Supplementary Material 1 (only available in the online version of this article) for further details on the CERC and assumptions used.
6. The baseline emissions trajectory as calculated by CERC is based on recent historical GDP and emissions intensity growth rates together with projections from IMF and IPCC for longer-term projections. The baseline thus differs from national estimates of a BAU trajectory.
7. See Supplementary Material 2 (only available in the online version of this article) for the results of this exercise.
8. Rand amounts were converted at the average exchange rate of 13.1 Rands to the US dollar – the average rate for July 2017 sourced from the South African Reserve Bank database (www.resbank.co.za).
9. Although the draft IRP 2016 is not officially adopted, it was chosen as a more accurate representation of the baseline than the official, but dated, IRP 2010. The CSIR work including updates to data on emissions abated and cost was prepared in response to the draft IRP 2016.
10. Note that power generation in 2016 was approximately 90% from coal, 4% from nuclear and 6% from renewable and other sources (CSIR, 2017).
11. Note that CSIR data on emissions abated and costs were available for each year for the scenarios with the exception of the Decarbonised Scenario where it was only available for 2016, 2025 and 2030. For the intervening years, emissions data were

thus interpolated using a smoothed curve. Cost data were interpolated based on maintaining the approximate additional cost associated with the Decarbonised Scenario relative to the Least Cost Scenario for the years where data were available.

12. Total mitigation costs (or savings) in a given year are driven by the costs (or savings) associated with the measures included which can be lumpy over time. For example, in the Low Carbon Scenario, relatively limited emissions reductions are achieved through power generation measures in 2025 which drives the overall cost of mitigation up. By 2030, cheaper power generation measures play a much more significant role in reducing emission which allows for overall savings.

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